

Certified Global Optimality for Packing Eight Disks of Radii $1, 2, \dots, 8$ in a Circle

Computer-assisted proof note — verification draft

9 October 2026

Abstract

We give a computer-assisted proof of the exact minimum circular-container radius for disks of radii $1, 2, \dots, 8$. The optimum $R_0 \in (16.22174667655772, 16.22174667655773)$ is the unique root of an explicit six-angle closure equation. The lower bound already follows from disks $3, \dots, 8$: a rational radial subdivision with 53 nodes exhausts their 60 cyclic orders, using angular negative cycles at 26 leaves and a radial monotonicity barrier at the remaining leaf. An exact eight-disk realization gives the matching upper bound. All certificate decisions are replayed using integer and rational arithmetic; the proof is not a formal verification in a theorem prover.

1 The exact optimum and its contact equation

For each $i = 1, \dots, 8$, let the disk of radius i have center $p_i \in \mathbb{R}^2$. A packing in a circular container of radius R satisfies

$$\|p_i\| + i \leq R, \quad \|p_i - p_j\| \geq i + j \quad (i \neq j). \quad (1)$$

Put $s_i = \|p_i\|$. The candidate configuration has a six-disk outer ring in cyclic order

$$\mathcal{C} = (4, 8, 3, 6, 5, 7), \quad E_{\mathcal{C}} = \{(4, 8), (8, 3), (3, 6), (6, 5), (5, 7), (7, 4)\}. \quad (2)$$

For two disks touching the container, their necessary central angle at mutual contact is

$$\beta_{ij}(R) = 2 \arcsin \sqrt{\frac{ij}{(R-i)(R-j)}} \quad (R > i + j). \quad (3)$$

Define the six-angle sum

$$\Phi(R) = \sum_{(i,j) \in E_{\mathcal{C}}} \beta_{ij}(R), \quad \Phi(R_0) = 2\pi. \quad (4)$$

Each β_{ij} is continuous and strictly decreasing in R on the interval of interest. Exact rational interval evaluations certify

$$\Phi(16.22174667655772) > 2\pi, \quad \Phi(16.22174667655773) < 2\pi. \quad (5)$$

Thus Equation (4) defines a unique root

$$16.22174667655772 < R_0 < 16.22174667655773.$$

The decimals in this paper denote exact rationals when used as interval endpoints or certificate values.

Theorem 1.1 (Exact global optimality). *For eight nonoverlapping disks with radii $1, 2, \dots, 8$, the minimum radius of a containing circle is the exact number R_0 defined in (4). In particular, there is an eight-disk packing at $R = R_0$, but no such packing at $R < R_0$.*

We prove the upper bound first, then exclude every smaller radius using the six largest disks.

2 An exact eight-disk realization

At $R = R_0$, place disk 4 at $(R_0 - 4, 0)$. For the cyclic order (2), define

$$p_i = (R_0 - i)(\cos \theta_i, \sin \theta_i), \quad \theta_4 = 0, \quad \theta_{c_{k+1}} = \theta_{c_k} + \beta_{c_k, c_{k+1}}(R_0). \quad (6)$$

The first five successive pairs are tangent by the cosine law, and the last pair (7, 4) is tangent by the closing equation $\Phi(R_0) = 2\pi$. All six centers are on the container wall. For the two remaining disks use the rational coordinates

$$p_1 = (-3, -14), \quad p_2 = (-3, -5/2). \quad (7)$$

A rational interval verifier encloses R_0 , every rotation cosine and sine in (6), and all resultant center coordinates. It proves strict separation for all $\binom{8}{2} - 6 = 22$ noncontact pairs and strict containment for disks 1 and 2. In particular, it certifies the squared inequalities

$$\|p_i - p_j\|^2 - (i + j)^2 > 0 \quad \text{for every noncontact pair}, \quad (R_0 - i)^2 - \|p_i\|^2 > 0 \quad (i = 1, 2).$$

The six listed contact equalities follow symbolically from (4). Hence the packing exists at the exact root: $R^* \leq R_0$.

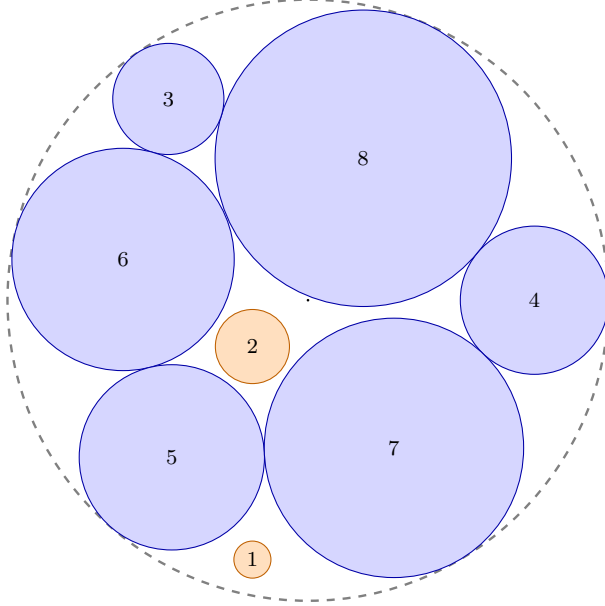


Figure 1: An illustrative drawing of the exact eight-disk witness. Blue disks comprise the six-disk tangent ring; orange disks have rational centers. Rounded coordinates are used for the drawing only, not for proof decisions.

3 Exact angular inequalities on radial rectangles

Assume for contradiction that a packing exists with $R < R_0$. It is contained in the rational cap

$$U = 16.22174667656 > R_0. \quad (8)$$

The six center radii s_i for $i \in \{3, 4, 5, 6, 7, 8\}$ satisfy

$$\ell_i := \max\left(0, \max_{1 \leq j \leq 8, j \neq i} (i + 2j - U)\right) \leq s_i \leq u_i := U - i. \quad (9)$$

Indeed, $i + j \leq \|p_i - p_j\| \leq s_i + s_j \leq s_i + (U - j)$. The six lower endpoints ℓ_i are strictly positive, so their center directions are defined.

For two disks with positive center radii x, y , put

$$c_{ij}(x, y) = \frac{x^2 + y^2 - (i + j)^2}{2xy}, \quad \phi_{ij}(x, y) = \arccos c_{ij}(x, y) \quad (10)$$

whenever the latter is defined. The cosine law and nonoverlap force the smaller central angle to be at least ϕ_{ij} . Consequently, in a counterclockwise cyclic order of center directions, each directed angular gap between two consecutive selected disks is at least their necessary smaller angle.

Lemma 3.1 (Four-corner cosine bound). *For $d > 0$, the function $c_d(x, y) = (x^2 + y^2 - d^2)/(2xy)$ attains its maximum on any positive rectangular domain at one of its four corners.*

Proof. For fixed $y > 0$,

$$\frac{\partial c_d}{\partial x} = \frac{x^2 - y^2 + d^2}{2yx^2}.$$

As x increases, the numerator strictly increases, so an interior stationary point, if any, is a minimum. Thus the x -maximum lies at an endpoint. Applying the same argument in y on these two edges establishes the claim. $\square$

Lemma 3.2 (Certified angular ticks). *Let B be a positive radial box and $q_{ij} \in \mathbb{Q} \cap [0, \pi]$. Suppose the rational Taylor polynomial*

$$C_{18}(q) = \sum_{k=0}^9 \frac{(-1)^k q^{2k}}{(2k)!}$$

satisfies

$$C_{18}(q_{ij}) \geq \max_{x \in \{\ell_i, u_i\}, y \in \{\ell_j, u_j\}} c_{ij}(x, y). \quad (11)$$

Then every feasible packing whose radial vector lies in B has pairwise smaller central angle at least q_{ij} .

Proof. For $q \in [0, \pi]$, the alternating Taylor remainder gives $\cos q \geq C_{18}(q)$. Lemma 3.1 bounds the radial cosine by the right-hand side of (11). For an actual nonoverlapping pair, $c_{ij}(s_i, s_j) \leq \cos q_{ij}$, so monotonicity of cosine on $[0, \pi]$ yields the assertion. $\square$

For a lifted cyclic order $(i_0, \dots, i_5)$, let $\theta_0 \leq \dots \leq \theta_5 < \theta_0 + 2\pi$ be the center directions. Any certified pair tick, including nonadjacent pairs, gives

$$q_{i_a i_b} \leq \theta_b - \theta_a \leq 2\pi - q_{i_a i_b} \quad (0 \leq a < b \leq 5). \quad (12)$$

The directed inequalities can be written as $\theta_v - \theta_u \leq w_{uv}$. Any directed cycle with negative total weight contradicts feasibility, since summing its inequalities yields $0 \leq \sum w_{uv} < 0$.

4 A complete rational exclusion tree

The six labeled disk directions have $(6-1)!/2 = 60$ cyclic orders up to rotation and reflection. The verifier enumerates every representative with disk 8 first and fixes one of the two mirror orientations.

The proof certificate recursively splits the initial rational box (9), propagating the necessary radial pair inequalities $s_i + s_j \geq i + j$. Each split is at a rational interior point, and its two closed children cover the parent domain. At each leaf the verifier uses Lemma 3.2 and exact difference-constraint graphs to exclude cyclic orders.

For positional indices $a < b$, it encodes (12) as the two directed edges

$$b \longrightarrow a : -q_{i_a i_b}, \quad a \longrightarrow b : 2\pi_+ - q_{i_a i_b}, \quad \pi_+ = 3.141592653590 > \pi. \quad (13)$$

The inequality $\pi < \pi_+$ is itself checked with Machin's identity and exact alternating rational arctangent bounds. Each edge weight is rational, and the verifier checks that a recorded negative cycle is indeed a simple directed cycle with the declared weights and strictly negative rational sum.

Proposition 4.1 (Exhaustive finite reduction). *A rational 53-node binary certificate covers every center-radius vector compatible with $R \leq U$. Its leaves comprise 26 angular leaves, each certifying a negative cycle for all 60 cyclic orders, and one remaining leaf that certifies negative cycles for 59 orders. The sole unexcluded cyclic order at that last leaf is*

$$(8, 3, 6, 5, 7, 4), \quad (14)$$

which is a rotation of the contact ring (2).

Proof. The provided checker replays all 26 rational splits and all 27 leaves, verifies parent–child coverage and the radial propagation rules, checks each supplied Taylor angle tick, and recomputes each recorded negative cycle using fractions. It verifies 1,619 order-specific cycles in total. As an example, in one ordinary leaf for order $(8, 3, 5, 7, 4, 6)$, the positional cycle corresponds to

$$6 \rightarrow 4 \rightarrow 7 \rightarrow 3 \rightarrow 8 \rightarrow 6.$$

Its five rational angle ticks are 1.138735, 1.379586, 1.231977, 1.490148, 1.713582; the cycle weight is

$$2\pi_+ - (1.138735 + 1.379586 + 1.231977 + 1.490148 + 1.713582) = -0.67084269282 < 0.$$

Every other leaf/order contradiction is represented by corresponding rational data in the accompanying JSON certificate. The rational replay rejects any unrecorded or inconsistent branch or cycle. $\square$

Table 1: Size of the replayed finite global certificate.

Recorded nodes	53
Rational splits	26
Leaves	27
Leaves excluding all 60 cyclic orders	26
Final leaves retaining one cyclic order	1
Explicit negative-cycle checks	1,619
Maximum subdivision depth	23

5 Closing the last order by wall-angle monotonicity

It remains to exclude the cyclic order (14) when $R < R_0$.

Lemma 5.1 (Radial wall comparison). *Let i, j be disks with center radii in positive intervals $[\ell_i, u_i]$, $[\ell_j, u_j]$. Suppose throughout the rectangle*

$$x + y > i + j, \quad |x - y| < i + j, \quad x^2 - y^2 + (i + j)^2 > 0, \quad y^2 - x^2 + (i + j)^2 > 0.$$

Then $\phi_{ij}(x, y)$ is strictly decreasing in each radial coordinate separately. In particular, if $x \leq R - i$ and $y \leq R - j$ and the rectangle includes these wall radii, then

$$\phi_{ij}(x, y) \geq \phi_{ij}(R - i, R - j) = \beta_{ij}(R). \quad (15)$$

Proof. Direct differentiation of (10) gives

$$\frac{\partial \phi_{ij}}{\partial x} = -\frac{x^2 - y^2 + (i + j)^2}{2x^2 y \sin \phi_{ij}(x, y)}. \quad (16)$$

The assumed triangle inequalities ensure $0 < \phi_{ij} < \pi$; thus the denominator is positive and the numerator is strictly positive. The corresponding derivative with respect to y is likewise negative. Increase the radial variables in the angle formula one at a time; this is an analytic comparison, not a physical displacement of disks. $\square$

For the last radial leaf, let ℓ'_i be its propagated lower endpoints. The exact checker verifies for each adjacent pair $(i, j) \in E_C$ that the extended rectangle

$$[\ell'_i, U - i] \times [\ell'_j, U - j]$$

satisfies the hypotheses of Lemma 5.1. In particular, it checks the endpoint conditions

$$\begin{aligned} \ell'_i + \ell'_j &> i + j, \\ \max\{|\ell'_i - (U - j)|, |\ell'_j - (U - i)|\} &< i + j, \\ (\ell'_i)^2 - (U - j)^2 + (i + j)^2 &> 0, \end{aligned}$$

and the last inequality with i and j exchanged. Table 2 gives readable decimal displays; all actual checks use the exact rational endpoints in the certificate.

Table 2: Final barrier-leaf radial ranges (decimal displays, not proof inputs).

Disk i	Certified lower endpoint ℓ'_i	Upper endpoint $U - i$
3	12.895387509275	13.221746676560
4	11.694028341990	12.221746676560
5	10.819028341990	11.221746676560
6	9.944028341990	10.221746676560
7	8.916310007420	9.221746676560
8	7.916310007420	8.221746676560

Now suppose a packing with $R < R_0$ survives the finite tree. Proposition 4.1 forces the cyclic order (14) and the last radial box. Lemma 5.1, applied to all six consecutive pairs, yields

$$\underbrace{2\pi}_{\text{complete turn}} \geq \sum_{(i,j) \in E_C} \phi_{ij}(s_i, s_j) \geq \sum_{(i,j) \in E_C} \beta_{ij}(R) > \sum_{(i,j) \in E_C} \beta_{ij}(R_0) = 2\pi. \quad (17)$$

The strict inequality follows because $R < R_0$ and each β_{ij} is strictly decreasing in R . This contradiction proves $R^* \geq R_0$; Section 2 established the reverse inequality. Theorem 1.1 follows.

6 Reproducibility and scope

The verifier files are

```
certify_integer_eight_global.py
verify_eight_upper.py
verify_eight_global_tree.py
eight_global_proof_tree.json
```

Run `python3 certify_integer_eight_global.py` from their directory with Python 3.11 or newer, without `-O/-OO`. The root/witness verifier checks rational enclosures of radicals, inverse trigonometric terms, π , and strict geometric slacks. The global-tree verifier checks all rational split and angle-cycle data and the final monotonicity conditions. Certificate discovery code is not required to replay the results.

Remark 6.1 (Trust boundary). *This is a computer-assisted mathematical proof conditional on the soundness of the accompanying verification code and Python’s exact integer and `Fraction` semantics. It is neither an end-to-end Lean formalization nor a claim of independent peer review. The exact value is specified by the isolated transcendental-looking angular closure equation; no minimal polynomial is claimed in this note.*